Chapter 6 · Solutions

P27 · Pack whole records

Hint. A block's leftover bytes cannot hold part of the next unspanned record.

Records per block=floor(1024/100)=10. Required blocks=ceil(101/10)=11. The shortcut ceil(10100/1024)=10 treats unused space as available across block boundaries, which is not allowed.

With headers, usable bytes=1024−50=974. Capacity=floor(974/100)=9. Required blocks=ceil(101/9)=12. Check: 11 blocks hold only 99 records.

P28 · Count entries before levels

Hint. Sparse anchors count data blocks; dense entries count records.

Data capacity is 10 records per block, so the file has 1000 blocks. An index block holds 1000/20=50 entries.

Layout	Lowest-level entries	Index blocks, bottom to top	Index levels	Lookup including data
sparse	1000	20,1	2	3 I/Os
dense	10,000	200,4,1	3	4 I/Os

The top has to fit in one block. Count one page at each index level, then the requested data page.

P29 · Write both byte budgets

Hint. An internal node has one more pointer than separator.

$4p+8(p-1)\le 1024$ gives $p\le 86$. At $p=86$ the usage is 1024; $p=87$ does not fit. Leaf entries satisfy $20q+4\le 1024$, giving $q=51$.

After doubling child pointers, $8p+8(p-1)\le 1024$ gives $16p\le 1032$, so $p=64$. The leaf record pointer and link did not change, so leaf capacity remains 51.

P30 · Build from the leaves

Hint. Round up at every parent layer.

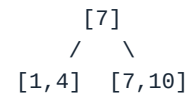
10,000 records need 200 leaves, then 2 parents, then 1 root. Height: 3 levels or 2 edges. 5000 records need 100 leaves and one root. Height: 2 levels or 1 edge.

For exactly 3 levels, the minimum is 2 root children ×50 leaves per internal child ×25 records per leaf=2500. The maximum is 100×100×50=500,000. Root occupancy and leaf capacity are different factors.

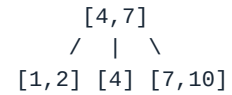
P31 · Repair each overflow or underflow

Hint. Copy a leaf separator. Promote an internal separator.

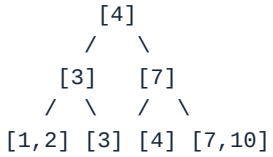
Initial tree:



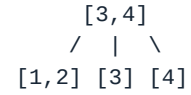
Insert 2. Leaf [1,2,4] splits into [1,2] and [4]; copy 4 to the root:



Insert 3. Leaf [1,2,3] splits into [1,2] and [3]. The root would be [3,4,7], with four children, so it overflows. Promote 4:



Delete 10. The last leaf becomes [7] and meets minimum occupancy. Delete 7. Its leaf becomes empty; [4] cannot lend, so merge them. The right internal node has only one child and underflows. The left internal node has only its minimum two children, so it cannot lend. Merge the internal nodes and collapse the root:



Check. All leaves stay at one depth. Each internal separator is the minimum of the right subtree. No record appears twice because of an internal separator.

P32 · Match the predicate and count pages

Hint. Hashing gives equality buckets; B+ order gives ranges.

Rating>3.4 uses B+. Genre='Comedy' and exact MovieID can use either standard hashing or B+ lookup.

A 3-level path with root cached reads two index pages. Add one data fetch: 3 I/Os. A covering lookup needs only 2.

For the range, the path's two reads already include the first leaf. Add the other four leaves and six data pages: 2+4+6=12 I/Os. Counting all five leaves again would double-count the first one.